VLM-DENTAL: synthetic trace synthesis and two-stage training pipeline

Public datasets



DENTEX + Tufts
expert labels (FDI)

Trace generation: LangGraph agent loop

frontier teacher via rate-limited provider pool
ground-truth-directed agent loop



reasoning turns interleaved with real tool output

8 tools: zoom_crop - window_level - denoise - enhance_contrast - contralateral_compare - locate_tooth - fdi_label - nudge_crop

tiered perturbation of the box shown to the teacher

Verification

independent, cross-family
verifier vs expert labels;
repair-and-re-verify loop
dual-gate directive audit

92 first-pass rejections,
3 measured failure modes

Verified corpus

3,694 traces = 1,847 + 1,847
two arms, exact 1:1 parity
5 cohorts, 2 institutions
44,350 real tool calls

0 directive leaks

Stage 1 -- SFT

QLoRA on Qwen3.5-9B

multi-turn trajectories
interleave the real crops
the tools produced;
loss masked to
assistant tokens

Stage 2 -- GRPO

ground-truth-anchored
composite reward:
multi-finding F1 accuracy,
format, tool validity,
complexity-scaled efficiency

locate_tooth and nudge_crop exempt from call cost

Policy

self-correcting
tool-use
agent

Eval

DENTEX-native
metrics,
ablations

One agent loop, one tool dispatcher, one image-input geometry (native, or canonical views when training on TPU) across trace generation, RL rollouts, and evaluation; tool calls execute for real against the source image at every stage.

During trace generation only: locate_tooth returns the expert-verified box for annotated teeth, and the perturbation hook independently perturbs the box shown.